Yelchuri Venkata Sai Harsha

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Education

R.V. College of Engineering, Bangalore, India — GPA: 9.01/10
B. E., Information Science and Engineering — Aug 2018 - Aug 2022

Publications

S=In Submission, C=Conference, W=Workshop, P=Poster/Demo, M=Manuscript in Preparation

M.1 **Robotwin: A Robotic Teleoperation Framework Using Digital Twins tasks**
Harsha Yelchuri, Nithish K Gnani, Diwakar Kumar Singh, T. V. Prabhakar

C.1 **SeMaScore: a new evaluation metric for automatic speech recognition tasks**
Zitha Sasindran, Harsha Yelchuri, T. V. Prabhakar
Conference of the International Speech Communication Association 2024 — **[Interspeech 2024]**

C.2 **Towards a resource-efficient semi-asynchronous federated learning for heterogeneous devices**
Zitha Sasindran, Harsha Yelchuri and T. V. Prabhakar
30th National Conference on Communications — **[NCC 2024]**

C.3 **Ed-Fed: A generic federated learning framework with resource-aware client selection for edge devices**
Zitha Sasindran, Harsha Yelchuri, T. V. Prabhakar
International Joint Conference on Neural Networks — **[IJCNN 2023]**

C.4 **H_eval: A new hybrid evaluation metric for automatic speech recognition tasks**
Zitha Sasindran, Harsha Yelchuri, Supreeth Rao, T. V. Prabhakar
Automatic Speech Recognition and Understanding — **[ASRU 2023]**

C.5 **MobileASR: A resource-aware on-device personalization framework for automatic speech recognition in mobile phones**
Zitha Sasindran, Harsha Yelchuri, Pooja Rao, T. V. Prabhakar
The Third International Conference on AI ML Systems — **[AI-ML Systems 2023]**

P.1 **SmartSV: Real-Time Solenoid Valve Diagnosis at Embedded Edge-Level**
Harsha Yelchuri, Shrutkirthi S. Godkhindi,, Prajwal BN, Vishwanath Shastry, T. V. Prabhakar
TinyML Asia Technical Forum 2023
16th International Conference on COMmunication Systems & NETworkS — **[COMSNETS 2024]**

P.2 **GASP: Gap Analysis for Spark Plug**
Harsha Yelchuri, Abhigna V Kumar, Siddharth Sahu, Savinay Kumar, Shrutkirthi S. Godkhindi, T. V. Prabhakar
16th International Conference on COMmunication Systems & NETworkS — **[COMSNETS 2024]**

S.1 **A review of TinyML**
Harsha Yelchuri, Rashmi R — **[2023]**

Research Experience

Post Baccalaureate Fellow, — Aug 2024 - Present
Indian Institute of Technology Madras, India

- Exploring the application of diffusion models in control for safe reinforcement learning settings.
- Investigating the issue of catastrophic forgetting in world models for long-horizon robotic tasks.

Project Associate, — Sep 2022 - Aug 2024
Indian Institute of Science, Bengaluru, India

1. **Robotics**
   - Designed and implemented an intelligent intercity (400 Km) teleoperation system.
   - Created a digital twin to mediate between the human operator and the physical robot (UR3).
   - Developed algorithms to update the digital twin in real-time regarding discrepancies arising in the remote environment.

2. **Evaluation Metrics using Large Language Models**
    - Developed two novel evaluation metrics to address the limitations of traditional Word Error Rate in speech-to-text models assessment.
    - The first metric H_{eval} is a weighted combination of semantic distance score and non-keyword error rate.
    - The second metric $SeMaScore$ employs a segment-wise mapping and scoring algorithm and leverages both the error rate and a more robust similarity score.
3. **Federated Learning**
    - Developed an end-to-end Federated Learning framework for edge devices in speech-to-text tasks.
    - Devised a novel client selection algorithm employing the bandit setting of Reinforcement Learning to bring forth fairness in client selection.
4. **On-Device Training**
    - Created a novel methodology for training large models on mobile devices by optimizing resource utilization based on mobile phone capabilities, such as RAM and battery percentage.

Research Intern, Feb 2022 - Sep 2022
Indian Institute of Science, Bengaluru, India

- Created an end-to-end machine learning pipeline for Predictive Maintenance, including fault detection and remaining useful life estimation of a solenoid valve.
- Successfully deployed trained models on edge device (Arduino Nano 33 BLE Sense) for real-time on-device inference.

Research Intern, Feb 2022 - Sep 2022
Samsung Research Institute, Bengaluru, India

- Devoloped a Bi-LSTM model based on textual input for a multi-class emotion classification (e.g., happy, sad, angry, and more).
- Expanded the project's scope to accommodate larger contexts (e.g., group of sentences).

Industry Experience

Machine Learning Lead, Jan 2023 - Present
EdgeSeanara, Bengaluru, India

- Currently directing the machine learning team for predictive maintenance of the fish tank.
- Currently leading the machine learning team on a proof of concept for fruit quality assessment.

Software Engineer Intern, Jun 2021 - Jul 2021
Boeing India Private Limited, Bengaluru, India

- Provided performance analytics to pilots using flight operational data as a part of pilot assessment training.
- Analyzed 1646 files of real-life flight data with each file consisting of 4196 parameters.
- Developed algorithms for the identification of two new Non-Flight Operations Quality Assurance events.

Teaching Experience

Teaching Assistant, Aug 2023 - Dec 2023
Indian Institute of Science, Bengaluru, India Aug 2022 - Dec 2022

- Assisted in conducting the lab sessions (in aspects of IoT protocols, machine learning, and micro-controllers), boot camps for the course "Design for Internet of Things" taught by Dr. T. V. Prabhakar.
- Guided students in the completion of mini-projects.

Guided Students

Soma Srija, Shreya Shahane Oct 2023 - Jan 2024
EdgeSeanara, Bengaluru, India
Project: Fruit Quality Inspection

- Developed deep learning models for fruit breed identification and quality inspection using texture analysis.

Siddharth Sahu, Savinay Kumar Aug 2023 - Dec 2023
Indian Institute of Science, Bengaluru, India
Project: Real-time identification of defective spark plugs on a conveyor belt

- Developed a system that uses a camera and a robotic arm to automatically inspect and sort spark plugs based on their gap length.

Kumar Raj Gummadi, Kaushik Raghunandan Hasabnis Aug 2022 - Dec 2022
Indian Institute of Science, Bengaluru, India
Project: Machine Vision based Foreign Object Detection

- Developed a Line scan camera-based machine vision solution to detect Foreign Objects on the surface.
- The system is capable of detecting rubber, stone, and metallic pieces of reasonable sizes of about 2mm.

Invited Talks & Conducted Boot Camps

Speaker: TinyML: Machine Learning on IoT Devices
The George Washington University, Washington, United States.

Boot Camp Instructor - TinyML
Indian Institute of Science, Bengaluru, India.

- Implementation of deep learning models on micro-controllers on tasks such as object recognition, image classification, and word recognition.
- Hands-on experience with hardware such as nrf52840 (Arduino nano 33 BLE), Arduino Nicla Voice & Arduino Nicla Vision.

Service & Achievements

Volunteer for eVidyaloka Trust. Created content in Maths and Science domain under the "Helping local schools to achieve good results and enhance their enrolment in Higher/ Technical/ Vocational education" vertical.

Volunteer for NSS. During my undergraduate studies in the pandemic, I volunteered over 200 hours to STEM education programs and environmental conservation initiatives.

Senior Member of Coding Club. Organized a variety of activities, including weekly coding competitions and hackathons, and also guided juniors towards their career objectives.

SmartX hackathon 2022. Our team secured 10th place out of 165 teams. As part of this hackathon, we designed a prototype that provides music and bright light therapy to the patients.

Open-Source Contributions

Ed-Fed Framework

- Contributed to the development of one of the first federated learning frameworks for mobile devices, with an emphasis on automatic speech recognition.

Technical Content Writer

- Initiated multiple blogs focused on disseminating knowledge about emerging technologies, with five published articles to date.

Skills

Tools and Technologies. Python, Tensorflow, PyTorch, MATLAB, ROS, Isaac Sim, Git, C, Java, Dart, HTML, CSS, JavaScript, NodeJS, Flask, MongoDB, Android Studio, Flutter.

Development Boards. Nano 33 BLE Sense, Nicla vision, Nicla voice, Raspberry Pi Pico, ESP32.

Robots. UR3, CrazyFly, Kobuki, DOBOT Magician.

Sensors. Line scan camera, 3D LiDAR, RADAR, Triad spectroscopy, Galvanic skin response.